

```

import cv2
import matplotlib.pyplot as plt
from google.colab import files

# Upload image
print("Upload a medical image or any grayscale image")
uploaded = files.upload()

filename = list(uploaded.keys())[0]

# Read image in grayscale
img = cv2.imread(filename, cv2.IMREAD_GRAYSCALE)

# -----
# Pixel Resolution
# -----

low_resolution = cv2.resize(img, (128,128), interpolation=cv2.INTER_AREA)
low_resolution = cv2.resize(low_resolution,
                             (img.shape[1], img.shape[0]),
                             interpolation=cv2.INTER_NEAREST)

# -----
# Intensity Resolution
# -----

# Reduce image to 16 gray levels (4-bit)
intensity_reduced = (img // 16) * 16

# -----
# Display Results
# -----

plt.figure(figsize=(15,5))

plt.subplot(1,3,1)
plt.imshow(img, cmap='gray')
plt.title("Original Image")
plt.axis("off")

plt.subplot(1,3,2)
plt.imshow(low_resolution, cmap='gray')
plt.title("Low Pixel Resolution")
plt.axis("off")

plt.subplot(1,3,3)
plt.imshow(intensity_reduced, cmap='gray')
plt.title("Low Intensity Resolution")
plt.axis("off")

plt.show()

# -----
# Explanation
# -----

print("\nObservation:")
print("1. Low Pixel Resolution:")
print("   - Image loses fine details.")
print("   - Edges become blocky or blurry.")

print("\n2. Low Intensity Resolution:")
print("   - Image has fewer gray levels.")
print("   - Smooth intensity transitions become banded.")

```

```
print("\nMedical Imaging:")  
print("- High Pixel Resolution helps detect small tumors, fractures, and blood vessels.")  
print("- High Intensity Resolution helps distinguish tissues with similar densities.")
```

Upload a medical image or any grayscale image

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Saving download.jpg to download.jpg

Original Image



Low Pixel Resolution



Low Intensity Resolution



Observation:

1. Low Pixel Resolution:
 - Image loses fine details.
 - Edges become blocky or blurry.
2. Low Intensity Resolution:
 - Image has fewer gray levels.
 - Smooth intensity transitions become banded.

Medical Imaging:

- High Pixel Resolution helps detect small tumors, fractures, and blood vessels.
- High Intensity Resolution helps distinguish tissues with similar densities.